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Philosophy of Science and Technologies examination – Paper 1

In this document, I discuss the topic of demarcation of science presented by Professor Giovanni Valente during the course of Philosophy of Science and Technologies, held at Politecnico di Milano during the academic year 2025/26.

The assignment pursued is the following : *The philosophical problem of demarcation between science and pseudo-science is closely related to the issues of scientific method and scientific progress. Explain in detail why that is the case. Then, describe the different solutions to the demarcation problem proposed by Inductivism, Falsificationism, Kuhn, Lakatos and Feyerabend, emphasizing merits and limitations of each of them. Finally, present your favorite view on the matter and illustrate it by applying to concrete cases (e.g. climate science, vaccination, creationism, etc.).*

Requirements: 3000 words

Words after this line: 3146

Demarcation of science:

The problem of demarcation of science comes from a collective human desire of classifying what knowledge is *scientific* and what is *not scientific*. Humans, both as individuals and as a collective, can collect knowledge through experience, however some knowledge may be considered more valid than other. Scientific knowledge is usually considered extremely valid, it can be used to perform predictions and it is usually believed as truthful. Scientific theories aim at developing a structured framework in which to use some specific laws to describe reality, they specify assumptions, quantities and levels of uncertainty. Scientific theories are considered to provide a trustful description of reality, through universal statements, such as laws, quantification of phenomena and causality relationships.

The problem arises not only when individual perception is different than common one, but also when common interpretation of experience is not actually leading to correct predictions. Then, one needs a recognized framework to judge which current of thoughts is “correct”, or at least to compare different ones to describe one as better than the others.

Scientific method and scientific progress:

In order to demark information given by science from other disciplines, one may look at two different aspects: the quality of such information, and the methodology with which it was obtained.

The first feature may regard its falsifiability through experiments or predictive success.

On the other side, the methodology is what, for each current of thought in philosophy of sciences, builds the *scientific method*. The information coming from such a method will be considered reliable; moreover, theories may be compared to decide which one is “more scientific”.

The composition of a scientific method, aims at prescribing what scientist ought to do, giving them rules; and, at the same time, aims at describing what scientists have been doing so far. Therefore, the focus rest on how science has developed in the past and a proper demarcation framework must be able to describe scientific progress and revolutions.

Observations and requirements:

Observations are linguistic statements about the state of affairs of the world. They come through the senses of individuals, and therefore are at first sight private and passive. If one aims at basing science on facts, this is a problem because privacy makes it subjective, while passiveness does not control which information is captured, leading to different interpretations. Scientific knowledge must be instead objective, that is “a common agreement for which a change of perspective does not change such agreement”.

Objectivity is the building block of reliable knowledge, it gives a level of trust to the information brought; it comes in two levels: objectivity of observations and of reasoning. This lead to the following requirements.

Firstly, one must conduct observations that are objective. That is, they are brought in a *careful and unprejudiced* way. In order to accommodate this requirement, communication is necessary. To guarantee communication however, repeatability is obligatory; one experiment must be repeatable by all of the scientific community.

Moreover, even for careful observers, observations are knowledge-dependent. One must also learn how to observe, an observer must be competent to actively isolate the proper information.

Secondly, the derivation of universal statements from observations must be objective. There must be a reasoning that is “sound”, that makes sense and brings agreement among the community.

To construct such sound reasoning, we make use of logic. In the course, two types of logic are studied: deductive and inductive.

Deductive reasoning:

Deductive reasoning transports truth from a particular statement to another particular statement. It works through implication. The logical form of a valid deduction argument is:

(All As are B) AND (X is A)

→ then X is B

This is said to be “truth preserving”, however its limitation is that it works only on a particular statement dealing with X, and not a class of elements from which one would like to derive a universal law.

Inductive reasoning:

Inductive reasoning, instead, expands knowledge from a finite number of instances to a class of phenomena, with infinite many instances. The logic form of a valid induction is, with n finite number:

(A₁ is B) AND (A₂ is B) AND... AND (A_n is B)

→ then all A_i are B (also for any i>n)

This holds if the three conditions are satisfied:

1. n must be large (enough)
2. Observations must be repeated under a big (enough) variety of conditions
3. There must be **no** case where an **acceptable** observation is conflicting with the statement proposed.

This can be rewritten through the *principle of induction*:

*If a large number of A's has been observed
under a wide variety of circumstances,
and if all those A's without exceptions possess the property B,
then all A's have the property B.*

Let's now discuss the main current of thoughts about demarcation of science.

1. Inductivism:

Method:

Inductivism uses inductive reasoning to derive universal statements from observational facts. It is an attempt to preserve the claim that “science is derived from facts”, by formalizing how one should treat facts.

Example:

In year 1005 b.c. the summer was warmer then the winter

In year 1004 b.c. the summer was warmer then the winter

...

In year 901 b.c. the summer was warmer then the winter.

→ then summer is always going to be warmer then the winter.

The example highlights how from a finite number of observations, one infers the behaviour of all the future years, generating a universal statement that should work for an infinite quantity of situations. That is, if induction principle is respected, then knowledge is expanded.

Inductivist science can be used both for explanations (“In year 2000 b.c. the summer was warmer then the winter because summers are warmer then the winters”) and for predictions (“next summer will be warmer then next winter”).

Any law derived from rigorous observations and rigorous application of induction principle is considered scientific.

Limitations:

The main problem with inductivism is circularity of inductive reasoning: inductive inference is considered to work only because it has only worked so far, therefore its use is justified by induction itself. This problem was described by David Hume, there is a main assumption taken in the use of induction which is that nature will behave in the future as it did in the past, however this assumption is not justified (if not by inductivism itself).

A solution to this problem is a *weak formulation of the induction principle* which concludes with “...then all *A*’s **probably** have the property *B*”. The circularity remains, although the conclusion becomes probabilistic rather than absolute, weakening the claim.

However, a second problem arises here on the concept of probability. How probable is a statement coming from weak induction principle? Probability is generally defined as “number of favourable outcomes” divided by “number of possible outcomes. Being the number of all possible future instances infinity, the probability of the law being true becomes zero (in the previous example, there will be infinitely many years to come).

Two other issues arise from inductivism. Firstly, the observations on which inductive inference is used do not comprehend the modern experiments involved in science, which are instead dealing with unobservable entities such as particles or molecules. Secondly, observations come with some non-removable error, while the universal laws are written as mathematical precise statements; how to deal with such uncertainty to justify the exactness of laws is not clear.

Lastly, inductivism views scientific progress as a cumulative process: increasing the observations, one increases the probability of the surviving laws being true. Science package grows of

“confirmed” or “probably true” universal statements. This does not explain properly scientific revolutions.

2. Falsificationism:

Method:

Falsificationism was ideated by Karl Popper. Inductive reasoning is replaced with deductive reasoning, and theories are considered scientific if they are falsifiable. This is the only necessary and sufficient condition to be considered scientific with respect to pseudo-scientific theories, then scientific progress will act on the theories to discard those not working (falsified) and keeping the good ones (not yet falsified, but falsifiable).

Example : Astrology results in being not scientific because usually its claims are so vague or flexible that no possible result counts against them.

The idea of Popper is characterized by the following points:

- i. No theory can be established as true, or even probably true, in light of observational evidence (inductive is not enough to claim some law to be true)
- ii. Only falsifiable hypotheses are scientific (necessary and sufficient condition)
- iii. Theories that fail to stand up observational and experimental tests ought to be eliminated (new experimental results may falsify previous scientific theories, not scientific anymore)
- iv. Only the “fittest” theory survives, that is the best available theory.
- v. Scientific practice proceeds by trial and error.
- vi. Scientists ought to follow a method based on *conjectures and refutations*.

This creates a dynamic picture of science, which depends on the previous state of art, where only the most suitable theory survive.

To choose which theory survives, or which one is the best at the moment, a sort of measure must be introduced: between two scientific (not yet falsified) theories, the best one is taken as the one which is more easily falsifiable.

Example: if a theory describes only gravitation on earth is less preferable than one describing gravitation in all the universe (more scenarios could falsify it).

3. Kuhn cycle of science:

Thomas Kuhn refused both inductivism and falsificationism, they fail to describe properly scientific revolution and do not represent the scientific practice. Kuhn observed that science happens in cycles characterized by the following steps:

Normal science → Anomaly → Crisis → Change in world view → Pre-paradigm phase → New paradigm → Normal science (on the new paradigm) → ...

Normal science

It lasts until all the scientists share a common *paradigm*. “Paradigm” is said to not be fully defined, however a paradigm can be characterized by a system of beliefs or model for solving problems.

Paradigm is the perspective from which scientists see research as a collective, it consists of fundamental laws, assumptions, ways of how to apply the laws, metaphysical principles and methodological prescriptions.

Example : in quantum mechanics one can mainly identify the paradigm with Planck's quantization and Schrödinger formalism, which are the achievements (triggered by the ultraviolet catastrophe, blackbody radiation anomalies, and the photoelectric effect).

According to Kuhn, once the new paradigm is established, there is no more real discovery. Every knowledge is already contained in the theoretical assumptions. During Normal Science, scientists keep working inside the recognized framework to solve problems through some rules descending from the paradigm itself, called "puzzles". However, this happens ignoring anomalies in the current paradigm, by justifying them as errors or experimental apparatus precision.

This view delimits as scientific only the problems that have solution inside such framework, even past ones are considered non-scientific or metaphysical.

Novelties

At some point in the progress, some theoretical or experimental results may challenge the current paradigm. Novelties of theories are called *inventions*, while novelties coming from experiments or observations are named *discoveries*. Discoveries happen when some experiments do not bring the expected results and the framework of science must be changed until it predicts correctly the results.

Example: the experiment of the photoelectric effect was wrongly predicted, the discovery was the behaviour of electric current, and the invention is that light comes in finite energy packets called *photons*.

Crisis

"The essential tension implicit in scientific research" is given as definition of *crisis*. This does not only include crisis which are necessary for paradigm shift, but also normal science puzzles can be seen as crisis. Crisis is indeed what drives scientific research, the "tension" pushing scientists to solve the inconsistencies.

Limitations:

According to Popper himself, his work was "much too schematic". It can describe but cannot predict when scientific revolutions will occur; some anomalies are later solved by scientists in the same cycle, while other times they may cause an acute crisis.

Moreover, because different currents of thought share different paradigms there is an *incommensurability* between different scientific theories, there is no way on how to compare them and therefore scientific revolutions happen when the scientific community shifts because they trust more the new paradigm. There seems to be no completely rational method for choosing among different theories.

4. Lakatos:

Structure of science:

Confines assumptions in *hard core* and *protective belt*. Hard core is constituted by those fundamental hypotheses that must not be changed, while the protective belt is the "outer layer" which is modified by research. The originality with respect to Kuhn's view, is that assumptions are spread among two different layers of importance, giving more flexibility.

Under this situation, science happens in *research programmes*: a group of central assumptions, which is taken as a dogma (for the moment), plus a set of *rules*.

The hard core can be shared among different scientific theories; many theories can evolve in parallel until one is accepted by the whole community.

The methodological rules guiding research are either *negative* or *positive heuristic*. Negative heuristic tells scientists what it is not permitted to remain in the research programme: that is to change the central assumptions. Positive heuristic guides scientists on how to perform research, what results to look for, which problems to investigate and which assumptions to modify.

Scientific progress:

Following from this description, scientific progress is observed by highlighting two possible types of research programmes: *progressive* and *degenerating*. Progressive when the research lead to novel predictions.

Example: Plank's quantization permitted to predict the "ultraviolet catastrophe", quantum mechanics in the beginning of XX century was a progressive research programme.

Degenerative is when the development is only protecting the hard core, not giving any novel prediction.

Revolutions fit in between the two: the time when science goes from a degenerative research to a progressive one. That happens when changes happen in the hard core of the degenerative research programme, leading scientists to adopt new fundamental assumptions (with the progressive research programme).

Many research programmes can be developed at the same time by different scientists, then a (non-relativistic) criterion is needed to compare them. The one developed by Lakatos was founded on two requirements: predictive power (novel predictions) and coherence (ability to guide future research).

Criticism :

Lakatos seems to describe science as irrational, scientists can stick with a research programme even if it is falsified at the moment. Research programmes can be compared rationally only over time.

Secondly, most of Lakatos idea is derived by the study of physical theories evolution, which could present some differences with other sciences in its mechanisms.

5. Feyerabend anarchy:

According to Feyerabend, when scientists develop a structured theory and argument with experiments they are only trying to convince other people by adding more sophisticated rhetorical arguments to their thesis. The abstract view of science doesn't differ from any other form of knowledge: people agree on its information. Science has no privilege role in knowledge, and it even does not differ from religion.

Example : *Energy medicine* are a set of theories that believe in healing the body by "channelling energy" into patients. This has no experimental and statistical evidence, however a community of people agree on its effectiveness, thus it is as scientific as medicine.

Feyerabend, arguments that Galileo didn't prove the *non-stationarity* of the earth, because he showed that it could be moving thanks to principle of Inertia but still he did not falsify the *stationarity* of it. Even if with a logical fallacy, he convinced people through *rhetorics*.

There seems to be no real method that scientists have followed, they are mainly pushed by intuition and then appeal to propaganda techniques to share their idea; and actually "Everywhere science is

enriched by unscientific methods and results“ (according to previous demarcations of science). Thus, he states scientists should not follow any rule, but instead follow creativity : *methodological anarchism*.

From this perspective, many theories considered scientific base themselves on different interpretations of the same observations. Thus, there is no rational basis for comparing different ones. Again we encounter the *incommensurability* of theories.

Criticism

Some theories seem to work more then others, they have more predictive success. So there must be some difference in knowledge, for which some theories are better.

Personal considerations

I personally admire falsificationism and the concept of “scientific is what is falsifiable”, however I think there are different levels of falsification of a theory.

Example: Phrenology affirmed that personality and mental abilities were related to skull’s shape. However this is today falsified and cannot enter as a scientific theory at the same level as Newtonian physics, they provide knowledge with different reliabilities.

Indeed, what I consider a core aspect of scientific theories is their *predictive success*, how much they can be used to predict observations and experimental results.

Lakatos view incorporates all of these concepts. Not only his method of comparison is comprehending predictive success, but also permits to older theories to still be scientific when the protective belt is modified.

Example: Newtonian physics can work in everyday life, and it can predict, one must consider it as part of sciences. However, it cannot work in presence of very strong gravitational field or at velocities close to lightspeed. Thus, the protective belt of assumptions can explain in which situations one can use Newtonian physics.

I see Lakatos’ view as accepting the complexity of making predictions, accepting that some theories may work not fully and that our human limits have not been able to provide us with a “Theory of everything”, but more importantly we probably will never find it.

In this situation I recognize Feyerabend’s point, science is deeply intersected with rhetorics and this infects its rationality. However, even if on the rhetorical level all theories may be the same, for their predictive success this is not working, and this is because phenomena in the world tend to follow rules (apparently). Actually, this irrationality is exactly the one that remains in Lakatos’ idea of science, where scientists will follow irrationally which research programme to follow.

I don’t think that retrospective judgement is a problem here, if one wants to be very precise and describe something as black or white (scientific or not), then one needs a lot of information which in this case coincide with future results that are not yet available. As inductivism can never prove that the variety of situations that were varied are enough, no one can judge if the causality relationships are fully understood, in the same way nobody will be able to fully judge scientific theories.

Example: continental drift was initially considered wrong because it lacked of a full description of the mechanisms of continents movement. However, it revealed to be correct later on... does this

make the theory non-scientific before it was approved? On one side yes, because it was not testable at the beginning, but actually it was true. The only point here is that one should give time to theories for development, and the judgement cannot arrive immediately.